

A Null-Diagonal Counterexample to Four Continuous Zauner Conjectures

Full negative answer for Conjectures 2.8, 2.9, 3.7, and 3.8 of arXiv:2209.12833

Automated proof attempt
agent_lane_03 model: GPT5.6

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Status: full counterexample, likely valid, pending human verification. All four conjectures are false as literally stated. On any infinite compact group with normalized Haar measure, the diagonal has product measure zero, whereas the conjectures require it to equal a strictly positive quantity.

Source statements

The source's Conjectures 2.8 and 2.9 (printed pp. 7–8) are respectively the continuous non-Archimedean functional and Hilbert-space Zauner conjectures. Conjectures 3.7 and 3.8 (printed pp. 14–15) are their p -adic analogues. Every one is quantified over a given locally compact group G and every dimension d , and every one includes the exact equality

$$|(\mu_G \times \mu_G)(\Delta)| = (\text{an off-diagonal correlation modulus}) = \frac{|\mu_G(G)|^2}{|d|}, \quad \Delta = \{(g, g) : g \in G\}. \quad (1)$$

The appended source excerpts reproduce the four statements and the decisive equalities exactly.

Theorem 1 (Simultaneous counterexample). *Let G be any infinite compact group with normalized Haar probability measure; in particular, take the circle group $G = \mathbb{T}$. Then none of Conjectures 2.8, 2.9, 3.7, or 3.8 is true for G . Each fails already in dimension $d = 1$.*

Proof intuition

The conjectures ask a continuously indexed family to be equiangular even on the diagonal at the level of measure: they identify the measure of the diagonal in $G \times G$ with the nonzero correlation constant. But Haar measure on an infinite compact group has no atoms. Consequently the diagonal is a product-null set. No choice of vectors, functionals, field, or frame operator can repair the resulting equation $0 = 1$ in dimension one.

Proof

Let μ be normalized Haar measure on an infinite compact group G . Every singleton has the same measure $a = \mu(\{e\})$. If $a > 0$, arbitrarily large finite sets would have measure Na , contradicting $\mu(G) = 1$. Thus $\mu(\{g\}) = 0$ for every $g \in G$. Tonelli's theorem now gives

$$(\mu \times \mu)(\Delta) = \int_G \mu(\{h : (g, h) \in \Delta\}) d\mu(g) = \int_G \mu(\{g\}) d\mu(g) = 0. \quad (2)$$

Set $d = 1$. Since $\mu(G) = 1$ and the absolute value of the multiplicative unit is one, the right-hand side of (1) is

$$\frac{|\mu(G)|^2}{|d|} = \frac{|1|^2}{|1|} = 1.$$

Equations (1) and (2) would therefore require $0 = 1$. This contradiction occurs before any of the vector, functional, integrability, tightness, or diagonalizability conditions are used, so it simultaneously refutes the functional and Hilbert-space versions.

For the non-Archimedean pair the hypothesis on the field is nonvacuous: for example, $\mathbb{K} = \mathbb{R}((t))$ with the t -adic absolute value satisfies the source's equation (1), because the lowest-order coefficients in a finite sum of squares of real Laurent series cannot cancel. For the p -adic pair use $\mathbb{Q}_p$, for any prime p . In either case $|1| = 1$, so the same contradiction applies. $\square$

Theorem 2 (Finite-group obstruction). *If a locally compact group G supports all the conditions in any one of the four conjectural lists (for even one dimension), then G is finite.*

Proof. In the functional formulations, choose a basis $(e_i)_{i=1}^d$ and its coordinate functionals (e_i^*) . Condition (i) makes each function $g \mapsto f_g(e_i)e_i^*(\tau_g)$ integrable, while self-normalization gives

$$1 = f_g(\tau_g) = \sum_{i=1}^d f_g(e_i)e_i^*(\tau_g).$$

Hence the constant function one is integrable and $\mu_G(G) < \infty$. The same argument in the Hilbert formulations uses $1 = \langle \tau_g, \tau_g \rangle$ and an orthonormal coordinate basis. A locally compact group has finite Haar measure exactly when it is compact.

The exact equality (1) has a nonzero right-hand side, so $(\mu_G \times \mu_G)(\Delta) > 0$. Haar invariance and Tonelli give

$$(\mu_G \times \mu_G)(\Delta) = \mu_G(G)\mu_G(\{e\}),$$

and thus $\mu_G(\{e\}) > 0$. As above, positive singleton mass is incompatible with infinitely many points in a compact neighborhood; consequently G is discrete. A compact discrete group is finite. $\square$

Scope and literature boundary

The theorem fully refutes the four universal conjectures. It does not fully classify Questions 2.6, 2.7, 3.5, and 3.6: those use a maximum of the diagonal mass and the off-diagonal correlation, so a null diagonal need not contradict their equality. A natural repair of the conjectures would restrict G to finite groups or remove the demand that the diagonal mass itself equal the correlation constant.

No exact-ID duplicate was found in the run indexes. Exact conjecture-name and arXiv searches through 11 August 2026 returned the source and its companion discrete papers, but no later correction or counterexample. This is a bounded novelty check, not an exhaustive priority claim. The nearby run packet for arXiv:2210.07062 concerns a discrete repeated-vector degeneracy; the present measure obstruction is distinct and specific to the continuous statements.

The script `code/verify_diagonal_measure.py` checks that normalized finite cyclic groups have diagonal product mass $1/N$, tending to the exact nonatomic value zero. The proof itself is the measure identity (2) and has no computational dependency.

References

- [1] K. Mahesh Krishna, *Continuous Non-Archimedean and p -adic Welch Bounds*, arXiv:2209.12833 (2022). Included as `source_paper.pdf`.

Exact source excerpt: non-Archimedean conjectures

$$\left(\sum_{g,h \in G, g \neq h} \dots \right) |d|$$

A particular case of Questions 2.6 and 2.7 are following continuous non-Archimedean versions of Zauner conjecture (see [2–5, 10–12, 33, 36, 43, 48, 51, 58, 66, 74, 88] for Zauner conjecture in Hilbert spaces, [53] for Zauner conjecture in Hilbert C^* -modules, [52] for Zauner conjecture in Banach spaces, [55] for Zauner conjecture in non-Archimedean Hilbert spaces, [56] for Zauner conjecture in p-adic Hilbert spaces and [54] for Zauner conjecture for non-Archimedean Banach spaces and p-adic Banach spaces).

Conjecture 2.8. (*Continuous Non-Archimedean Functional Zauner Conjecture*) *Let $\mathbb{K}$ non-Archimedean field satisfying Equation (1). Given locally compact group G and for each $d \in \mathbb{N}$, there exist a collection $\{\tau_g\}_{g \in G}$ in $\mathbb{K}^d$ (w.r.t. any non-Archimedean norm) and a collection $\{f_g\}_{g \in G}$ in $(\mathbb{K}^d)^*$ satisfying the following.*

- (i) *For every $x \in \mathbb{K}^d$ and for every $\phi \in (\mathbb{K}^d)^*$, the map*

$$G \ni g \mapsto f_g(x)\phi(\tau_g) \in \mathbb{K}^d$$

is measurable and integrable.

7

- (ii) *The map*

$$S_{f,\tau} : \mathbb{K}^d \ni x \mapsto \int_G f_g(x)\tau_g d\mu_G(g) \in \mathbb{K}^d$$

is a well-defined bounded linear operator.

- (iii) *The operator $S_{f,\tau}$ is diagonalizable.*

- (iv)

$$\int_{G \times G} |f_g(\tau_h)f_h(\tau_g)| d(\mu_G \times \mu_G)(g,h) < \infty.$$

- (v) $f_g(\tau_g) = 1$ *for all* $g \in G$.

- (vi)

$$|(\mu_G \times \mu_G)(\Delta)| = |f_g(\tau_h)f_h(\tau_g)| = \frac{|\mu(G)|^2}{|d|}, \quad \forall g, h \in G, g \neq h.$$

- (vii) $\|f_g\| = 1, \|\tau_g\| = 1$ *for all* $g \in G$.

Conjecture 2.9. (*Continuous Non-Archimedean Zauner Conjecture*) *Let $\mathbb{K}$ non-Archimedean field satisfying Equation (1). Given locally compact group G and for each $d \in \mathbb{N}$, there exists a collection $\{\tau_g\}_{g \in G}$ in $\mathbb{K}^d$ satisfying the following.*

- (i) *For every $x \in \mathbb{K}^d$ and for every $\phi \in (\mathbb{K}^d)^*$, the map*

$$G \ni g \mapsto \langle x, \tau_g \rangle \phi(\tau_g) \in \mathbb{K}^d$$

is measurable and integrable.

- (ii) *The map*

$$S_\tau : \mathbb{K}^d \ni x \mapsto \int_G \langle x, \tau_g \rangle \tau_g d\mu_G(g) \in \mathbb{K}^d$$

is a well-defined bounded linear operator.

- (iii) *The operator S_τ is diagonalizable.*

- (iv)

$$\int_{G \times G} |\langle \tau_g, \tau_h \rangle|^2 d(\mu_G \times \mu_G)(g,h) < \infty.$$

- (v) $\langle \tau_g, \tau_g \rangle = 1$ *for all* $g \in G$.

- (vi)

$$|(\mu_G \times \mu_G)(\Delta)| = |\langle \tau_g, \tau_h \rangle|^2 = \frac{|\mu(G)|^2}{|d|}, \quad \forall g, h \in G, g \neq h.$$

There are four bounds which are companions of Welch bounds in Hilbert spaces. To recall them we need the notion of Gerzon's bound.

Definition 2.10. [45] *Given $d \in \mathbb{N}$, define Gerzon's bound*

$$\mathcal{Z}(d, \mathbb{K}) := \begin{cases} d^2 & \text{if } \mathbb{K} = \mathbb{C} \\ \frac{d(d+1)}{2} & \text{if } \mathbb{K} = \mathbb{R}. \end{cases}$$

Theorem 2.11. [16, 22, 41, 45, 61, 65, 75, 85] *Define $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$ and $m := \dim_{\mathbb{R}}(\mathbb{K})/2$. If $\{\tau_j\}_{j=1}^n$ is any collection of unit vectors in $\mathbb{K}^d$, then*

8

Exact source excerpt: p -adic conjectures

A particular case of Questions 3.5 and 3.6 are following continuous p -adic Zauner conjectures.

Conjecture 3.7. (Continuous p -adic Functional Zauner Conjecture) *Let p be a prime. Given locally compact group G and for each $d \in \mathbb{N}$, there exist a collection $\{\tau_g\}_{g \in G}$ in $\mathbb{Q}_p^d$ (w.r.t. any non-Archimedean norm) and a collection $\{f_g\}_{g \in G}$ in $(\mathbb{Q}_p^d)^*$ satisfying the following.*

- (i) *For every $x \in \mathbb{Q}_p^d$ and for every $\phi \in (\mathbb{Q}_p^d)^*$, the map*

$$G \ni g \mapsto f_g(x)\phi(\tau_g) \in \mathbb{Q}_p^d$$

is measurable and integrable.

- (ii) *There exists $b \in \mathbb{Q}_p$ such that*

$$\int_G f_g(x)\tau_g d\mu_G(g) = bx, \quad \forall x \in \mathbb{Q}_p^d.$$

- (iii)

$$\int_{G \times G} |f_g(\tau_h)f_h(\tau_g)| d(\mu_G \times \mu_G)(g, h) < \infty.$$

- (iv) $f_g(\tau_g) = 1$ for all $g \in G$.

14

- (v)

$$|(\mu_G \times \mu_G)(\Delta)| = |f_g(\tau_h)f_h(\tau_g)| = \frac{|\mu(G)|^2}{|d|}, \quad \forall g, h \in G, g \neq h.$$

- (vi) $\|f_g\| = 1, \|\tau_g\| = 1$ for all $g \in G$.

Conjecture 3.8. (Continuous p -adic Zauner Conjecture) *Let p be a prime. Given locally compact group G and for each $d \in \mathbb{N}$, there exists a collection $\{\tau_g\}_{g \in G}$ in $\mathbb{Q}_p^d$ satisfying the following.*

- (i) *For every $x \in \mathbb{Q}_p^d$ and for every $\phi \in (\mathbb{Q}_p^d)^*$, the map*

$$G \ni g \mapsto \langle x, \tau_g \rangle \phi(\tau_g) \in \mathbb{Q}_p^d$$

is measurable and integrable.

- (ii) *There exists $b \in \mathbb{Q}_p$ such that*

$$\int_G \langle x, \tau_g \rangle \tau_g d\mu_G(g) = bx, \quad \forall x \in \mathbb{Q}_p^d.$$

- (iii)

$$\int_{G \times G} |\langle \tau_g, \tau_h \rangle|^2 d(\mu_G \times \mu_G)(g, h) < \infty.$$

- (iv) $\langle \tau_g, \tau_g \rangle = 1$ for all $g \in G$.

- (v)

$$|(\mu_G \times \mu_G)(\Delta)| = |\langle \tau_g, \tau_h \rangle|^2 = \frac{|\mu(G)|^2}{|d|}, \quad \forall g, h \in G, g \neq h.$$

- (vi) $\|\tau_g\| = 1$ for all $g \in G$.

Theorem 2.11, Theorem 3.1 and Corollary 3.2 give the following problems.

Question 3.9. Whether there is a continuous p -adic functional version of Theorem 2.11? In particular, does there exists a version of

- (i) *continuous p -adic functional Bukh-Cox bound?*
(ii) *continuous p -adic functional Orthoplex/Rankin bound?*